

v7.5 Bivariate Distribution Dominance - Visual Reporting Specification

Purpose

The report identifies where and when each fitted joint probability model dominates after model selection. It is explicitly post-fit: no bivariate family is assumed a priori.

Required selection field

best_model_code(doy, latitude, longitude) plus a model_names mapping in NetCDF metadata. A model code is the winning joint model for that climatological day and grid cell.

Visual products

Product	Purpose
Monthly dominance share	Fraction of grid-day cells won by each joint model in each calendar month.
DOY x model heatmap	Shows how model dominance changes through the climatological year.
Monthly spatial maps	Shows the dominant joint model at each grid cell for every month.
Model-code legend	Keeps model labels auditable and versioned.

Scientific interpretation

The dominant map is not a statement that one distribution is globally correct. It is a visualization of the model selected under the frozen fitting and scoring contract for each DOY and location. Empirical 2-D probability remains the non-parametric reference product; Gaussian copula, t-copula, Beta marginals, Skew-Normal, Pearson Type III, and Bimodal Normal components are candidates subject to fit diagnostics.

Temporal window

Distribution fitting uses a centered five-day window (target day ± 2 days) at the hourly level. The 1980-12-30/31 edge file is used only to complete the first target windows of 1981; 2021 data provide end-of-period padding. Padding observations do not become independent climatological targets.

Failure and provenance rule

The visual report is generated only from a validated model-selection NetCDF. Model codes, configuration hash, source checksum, input manifest, fitting settings, window rule, and selection rule must remain linked to the report.